Dr. Alexander Maximilian von Humboldt

Lead Distributed Systems Architect

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linkedin.com/in/dr-alexander-maximilian-von-humboldt-cloud-architect-2026 | github.com/alexander-maximilian-systems/distributed-consensus-raft-implementation

alexander-maximilian-portfolio-and-research-showcase.engineering.dev |

leetcode.com/u/alexander_maximilian_systems?tab=profile&view=detailed

Professional Summary

Principal systems architect with 12+ years designing distributed consensus algorithms, high-throughput financial engines, and fault-tolerant cloud backbones. Author of peer-reviewed papers on distributed transaction scheduling.

Professional Experience

Principal Systems Architect — Apex Distributed Financial Systems Jan 2020 – Present

Berlin, Germany | Full time

- Architected Raft consensus engine in Rust processing 250,000 distributed orders per second.
- Engineered eBPF network packet filter reducing inter-node packet serialization latency by 45%.
- Eliminated distributed deadlock vulnerability during partition recovery through formal TLA+ verification.

Staff Infrastructure Engineer — Berlin Cloud Computing Labs Apr 2016 – Dec 2019

Berlin, Germany | Full time

- Engineered global cluster scheduling service orchestrating 4,000 bare-metal compute nodes.
- Published technical whitepapers on fault tolerance and split-brain resolution protocols.

Technical Projects

Formal Raft Verification Framework **GitHub**

Model checker and simulation engine for Raft consensus.

Rust, TLA+, Distributed Systems

- Implemented state-machine safety proof verification for randomized leader election cycles.
- Simulated 10,000 network partitions discovering zero safety violations or split-brain states.

Zero-Copy IPC Messaging Bus **GitHub**

Ultra-low-latency inter-process communication queue in C++.

C++20, Linux Shared Memory

- Designed ring-buffer messaging bus using shared memory and lock-free atomic primitives.
- Benchmarked round-trip latency achieving 42 nanoseconds on modern x86-64 hardware.

Technical Skills

Skills: Rust, Go, C++20, Distributed Consensus, Raft, Paxos, RocksDB, Kafka, PostgreSQL, Kubernetes, Linux Kernel eBPF, Terraform, AWS

Education

Doctor of Philosophy (Ph.D.) in Distributed Computer Systems Oct 2012 – Feb 2016

Humboldt University of Berlin | Berlin, Germany

Grade: Summa Cum Laude

Languages

German (Native), English (Fluent)